

**CLASSROOM CONTACT PROGRAMME**
(Academic Session : 2023-2024)

NEET(UG)

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PRE-MEDICAL : ENTHUSIAST ADVANCE COURSE PHASE : MED-I (B)

This Booklet contains 28 pages.

ASSIGNMENT
(FOR SELF EVALUATION)

Name of the Student (in Capitals) : _____

Form Number : in figures _____

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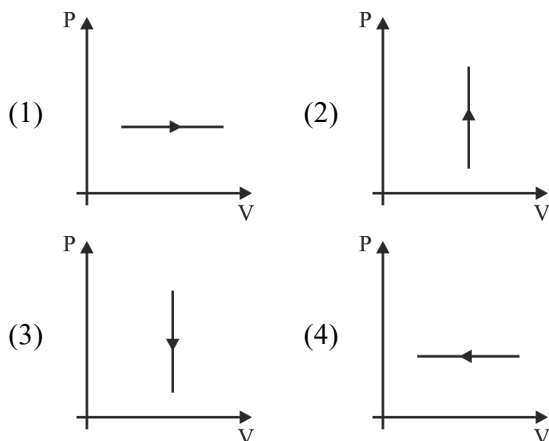
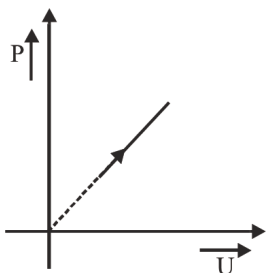
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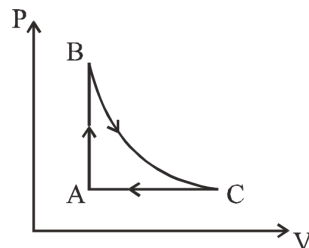
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SECTION-A (PHYSICS)

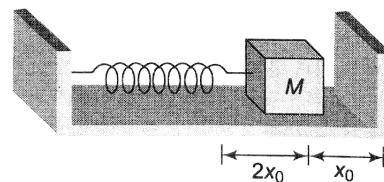
- Phase of a substance solid, liquid or gas is defined by :-
 - By temperature only
 - By pressure only
 - By both pressure & temperature
 - None
- Suppose ideal gas equation follows $VP^3 = \text{constant}$. Initial temperature and volume of the gas are T and V respectively. If gas expand to $27V$ then its temperature will become :-
 - T
 - $9T$
 - $27T$
 - $T/9$
- The given $P-U$ graph shows the variation of internal energy of an ideal gas with increase in pressure. Which of the following pressure-volume graph is equivalent to this graph?



- A sample of an ideal gas is taken through a cycle as shown in figure. It absorbs 50J of energy during the process AB, no heat during BC, rejects 70J during CA. 40J of work is done by the gas during BC. Internal energy of gas at A is 1500J, then work done on gas in CA process would be :-

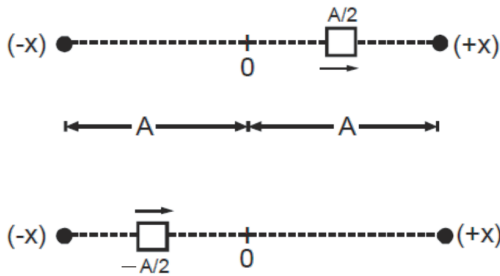


- 90 J (2) 20 J (3) 60 J (4) 70 J
- An iron block cools from 100°C to 90°C in time t_1 ; 90°C to 80°C in time t_2 and 80°C to 70°C in time t_3 . Which of the following relations is correct
 - $t_1 < t_2 < t_3$
 - $t_1 > t_2 > t_3$
 - $t_1 = t_2 = t_3$
 - $t_3 = \frac{t_1 + t_2}{2}$
 - One end of a spring of force constant k is fixed to a vertical wall and the other to a block of mass m resting on a smooth horizontal surface. There is another wall at a distance x_0 from the block. The spring is then compressed by $2x_0$ and released. The time taken to strike the wall is



- $\frac{1}{6} \pi \sqrt{\frac{k}{m}}$
- $\sqrt{\frac{k}{m}}$
- $\frac{2\pi}{3} \sqrt{\frac{m}{k}}$
- $\frac{\pi}{4} \sqrt{\frac{k}{m}}$

7. Two bodies performing S.H.M. have same amplitude and frequency. Their phases at a certain instant are as shown in the figure. The phase difference between them is



- (1) $\frac{11}{6}\pi$ (2) π (3) $\frac{\pi}{3}$ (4) $\frac{3}{5}\pi$
8. A particle is moving on x-axis has potential energy $U = 2 - 20x + 5x^2$ joule along x-axis. The particle is released at $x = -3$. The maximum value of 'x' will be [x is in meter and U is in joule] :-

- (1) 5 m (2) 3 m
(3) 7 m (4) 8 m

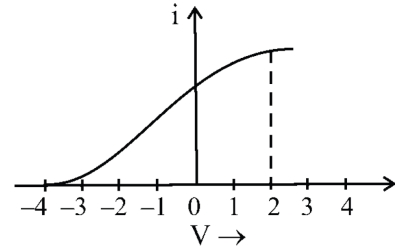
9. When metal surface is illuminated with light of wavelength λ , the stopping potential is V. When the same surface is illuminated by light of wavelength 2λ , the stopping potential is $V/3$, Threshold wavelength for metallic surface is :-

- (1) $4\lambda/3$
(2) 4λ
(3) 6λ
(4) $8\lambda/3$

10. When a monochromatic source of light is at a distance of 0.2 m from a photoelectric cell, the cut-off voltage and the saturation current are respectively 0.6 V and 18 mA. If the same source is placed 0.6 m away from the cell, then :

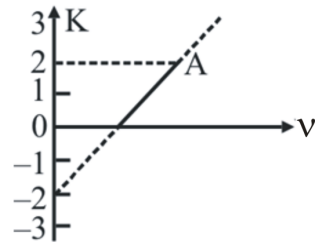
- (1) the stopping potential will be 0.2 V
(2) the stopping potential will be 1.8 V
(3) the saturation current will be 6.0 mA
(4) the saturation current will be 2.0 mA

11. Figure represents the graph of photo current I versus applied voltage (V). The maximum kinetic energy of the emitted photoelectrons is :-



- (1) 2 eV (2) 4 eV (3) 0 eV (4) 4 J

12. Figure represents a graph of maximum kinetic energy (K) of photoelectrons (in eV) and frequency (ν) for a metal used as cathode in photoelectric experiment. What is value of frequency at point A.

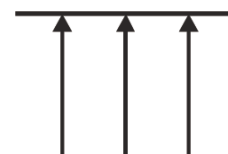


- (1) 5×10^{14} Hz (2) 10^{15} Hz
(3) 5×10^{15} Hz (4) 10^{16} Hz

13. Two photons having :-

- (1) Equal wavelength have equal linear momenta
(2) Equal energies have equal linear momenta
(3) Equal frequencies have equal linear momenta
(4) Equal linear momenta have equal wavelength

14. A plate of mass 10 g is in equilibrium in air due to force exerted by a light beam on the plate. Calculate power of the beam. Assume that the plate is perfectly absorbing.

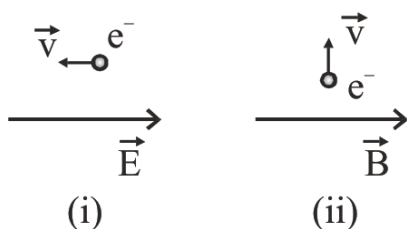


- (1) 1.5×10^7 W (2) 14×10^7 W
(3) 3×10^7 W (4) 8×10^7 W

15. In an experiment of photoelectric effect the stopping potential was measured to be V_1 and V_2 volts with incident light of wavelength λ and $\lambda/2$ respectively. The relation between V_1 and V_2 may be :-

- (1) $V_2 < V_1$ (2) $V_1 < V_2 < 2V_1$
(3) $V_2 = 2V_1$ (4) $V_2 > 2V_1$

16. An electron is moving through a field. It is moving (i) opposite an electric field (ii) perpendicular to a magnetic field as shown. For each situation the de-Broglie wave length of electron.



- (1) Increasing, increasing
(2) Increasing, decreasing
(3) Decreasing, same
(4) Same, same

17. Kinetic energy of an electron is 1600 eV. Find its de-Broglie wavelength :-

- (1) $0.3 \text{ \AA}$ (2) $0.15 \text{ \AA}$
(3) $8 \text{ \AA}$ (4) None of these

18. The maximum velocity of an electron emitted by light of wavelength λ incident on the surface of a metal of work function ϕ , is :-

- (1) $\left[\frac{2(hc + \lambda\phi)}{m\lambda} \right]^{1/2}$
(2) $\frac{2(hc - \lambda\phi)}{m}$
(3) $\left[\frac{2(hc - \lambda\phi)}{m\lambda} \right]^{1/2}$
(4) $\left[\frac{2(h\lambda - \phi)}{m} \right]^{1/2}$

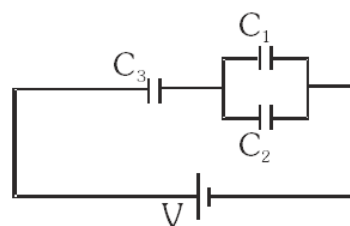
19. The electric field between the plates of a parallel plate capacitor when connected to a certain battery is E_0 . If the space between the plates of the capacitor is filled by introducing a material of dielectric constant K without disturbing the battery connections; the field between the plates will be :

- (1) KE_0 (2) E_0
(3) $\frac{E_0}{K}$ (4) None of the above

20. A slab of copper of thickness $\frac{d}{2}$ is introduced between the plates of a parallel plate capacitor where d is the separation between its two plates. If the capacitance of the capacitor without the copper slab is C and with copper slab is C' then $\frac{C'}{C}$ is :-

- (1) $\sqrt{2}$ (2) 2 (3) 1 (4) $\frac{1}{\sqrt{2}}$

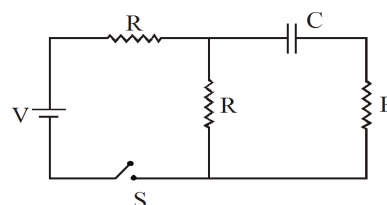
- 21.



Potential difference across C_3 is:

- (1) $\frac{(C_1 - C_2)V}{C_1 + C_2 + C_3}$ (2) $\frac{(C_1 + C_2)V}{C_1 + C_2 + C_3}$
(3) $\frac{(C_1 + 2C_2)V}{C_1 + C_2 + C_3}$ (4) $\frac{(C_1 - 2C_2)V}{C_1 + C_2 + C_3}$

22. Current drawn by the battery just after switch 'S' is closed is :-



- (1) $\frac{2V}{3R}$ (2) $\frac{3V}{2R}$
(3) $\frac{2V}{R}$ (4) $\frac{V}{R}$

23. Which of the following radiations are correctly in descending order of frequencies :-

- (1) Microwaves, Radio waves, X-Ray, Red light
- (2) X-Ray, Red light, Microwaves, Radio waves
- (3) X-Ray, Radio waves, Red light, Microwaves
- (4) Red light, Microwaves, X-Ray, Radio waves

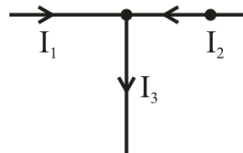
24. A point source of power 15 W is placed at a certain point in the space. The amplitude of magnetic field at a distance of 2 meter from source is :

- (1) 5×10^{-8} T (2) 6×10^{-8} T
- (3) 10×10^{-8} T (4) 1×10^{-8} T

25. For a plane electromagnetic wave direction of electric field (E) and magnetic field (B) is along $-\hat{i} + \hat{j}$ and $-\hat{i} - \hat{j}$ respectively. The direction of propagation of plane electromagnetic wave is :-

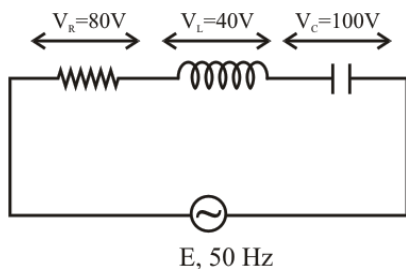
- (1) +x direction (2) +z direction
- (3) +y direction (4) -x direction

26. If current $I_1 = 3A \sin \omega t$ and $I_2 = 4A \cos \omega t$, then I_3 is :-



- (1) $5A \sin (\omega t + 53^\circ)$ (2) $5A \sin (\omega t + 37^\circ)$
- (3) $5A \sin (\omega t + 45^\circ)$ (4) $5A \sin (\omega t + 30^\circ)$

27. The value of alternating emf E in the given circuit will be:



- (1) 220 V (2) 140 V
- (3) 100 V (4) 20 V

28. The voltage of an ac source varies with time according to the equation

$$V = 100 \sin(100\pi t) \cos(100\pi t)$$

where t is in seconds and V is in volts. Then

- (1) The peak voltage of the source is 100 volts
- (2) The peak voltage of the source is 50 volts
- (3) The peak voltage of the source is $100/\sqrt{2}$ volts
- (4) The frequency of the source is 50 Hz

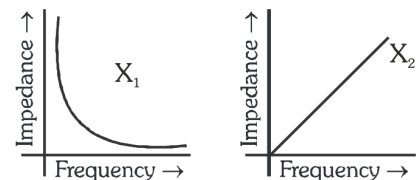
29. An alternating current is given by the equation $i = i_1 \cos \omega t + i_2 \sin \omega t$. R.M.S. current is given by

- (1) $\frac{1}{\sqrt{2}}(i_1 + i_2)$ (2) $\frac{1}{\sqrt{2}}(i_1 + i_2)^2$
- (3) $\frac{1}{\sqrt{2}}(i_1^2 + i_2^2)^{1/2}$ (4) $\frac{1}{2}(i_1^2 + i_2^2)^{1/2}$

30. In a series LCR circuit, resistance $R = 10\Omega$ and the impedance $Z = 20\Omega$. The phase difference between the current and the voltage is

- (1) 30°
- (2) 45°
- (3) 60°
- (4) 90°

31. The graphs given below depict the dependence of two reactive impedances X_1 and X_2 on the frequency of the alternating e.m.f. applied individually to them. We can then say that :



- (1) X_1 is an inductor and X_2 is a capacitor
- (2) X_1 is a resistor and X_2 is a capacitor
- (3) X_1 is a capacitor and X_2 is an inductor
- (4) X_1 is an inductor and X_2 is a resistor

32. A small square loop of wire of side ℓ is placed inside a large square loop of wire of side L ($L \gg \ell$). The loops are coplanar and their centres coincide. The mutual inductance of the system is proportional to :

- (1) ℓ/L (2) ℓ^2/L
(3) L/ℓ (4) L^2/ℓ

33. A transformer is used to light 140 watt 24 volt lamp from 240 volt AC mains; the current in the main cable is 0.7 amp. The efficiency of the transformer is :

- (1) 63.8% (2) 84%
(3) 83.3% (4) 48%

34. A step up transformer has turn ratio 10:1. A D.C. source of emf. 2 volt is fed to the primary; secondary voltage developed will be :-

- (1) 20 V (2) 10 V
(3) 2 V (4) zero

35. The core of the transformer is laminated to

- (1) Increase the secondary voltage
(2) Reduce eddy current loss due to eddy current
(3) Make it light weight
(4) To decrease cost of manufacture

SECTION-B (PHYSICS)

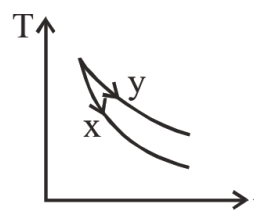
36. 10 gm of ice at -20°C is dropped into a calorimeter containing 10 gm of water at 10°C ; the specific heat of water is twice that of ice. When equilibrium is reached, the calorimeter will contain :-

- (1) 20 gm of water
(2) 20 gm of ice
(3) 10 gm ice and 10 gm water
(4) 5 gm ice and 15 gm water

37. A carnot engine has efficiency $1/5$. Efficiency becomes $1/3$ when temperature of sink is decreased by 50K. What is the temperature of sink?

- (1) 325 K (2) 375 K (3) 300 K (4) 350 K

38. The graph. Shown in the adjacent diagram, represents the variation of temperature (T) of two bodies, x and y having same surface area, with time(t) due to the emission of radiation. Find the correct relation between the emissivity (e) and absorptivity (a) of the two bodies :-



- (1) $e_x > e_y$ and $a_x < a_y$ (2) $e_x < e_y$ and $a_x > a_y$
(3) $e_x > e_y$ and $a_x > a_y$ (4) $e_x < e_y$ and $a_x < a_y$

39. A simple pendulum is executing simple harmonic motion with a time period T . If the length of the pendulum is increased by 21%, the increase in the time period of the pendulum of increased length is

- (1) 10% (2) 21% (3) 30% (4) 50%

40. When a metal is irradiated by monochromatic light, the maximum kinetic energy of the photo-electrons is 1.2eV. If frequency of the light is increased by 50% then maximum kinetic energy of photo-electron is 3.6eV. Evaluate the workfunction of the metal:-

- (1) 3.6 eV (2) 5.8 eV
(3) 9.7 eV (4) None of these

41. Light of wavelength $5000 \text{ \AA}$ falling on a sensitive surface. If the surface has received 10^{-7} J of energy, then the number of photons falling on the surface will be -

- (1) 5×10^{11} (2) 2.5×10^{11}
(3) 3×10^{11} (4) None of these

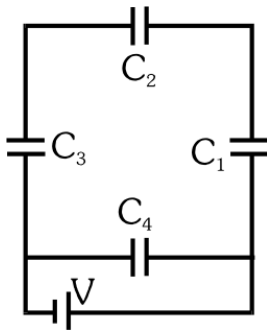
42. λ_e , λ_p and λ_α are the de-Broglie wavelength of electron, proton and α -particle, If all are accelerated by same potential, then

- (1) $\lambda_e < \lambda_p < \lambda_\alpha$ (2) $\lambda_e < \lambda_p > \lambda_\alpha$
 (3) $\lambda_e > \lambda_p > \lambda_\alpha$ (4) $\lambda_e = \lambda_p > \lambda_\alpha$

43. A proton has kinetic energy $E = 100$ keV which is equal to that of a photon. The wavelength of photon is λ_2 and that of proton is λ_1 . The ratio of λ_2/λ_1 is proportional to :-

- (1) E^2 (2) $E^{-1/2}$ (3) E^{-1} (4) $E^{1/2}$

44. A network of four capacitors of capacity equal to $C_1=C$, $C_2=2C$, $C_3=3C$ and $C_4=4C$ are connected to a battery as shown in the figure. The ratio of the charges on C_2 and C_4 is :-

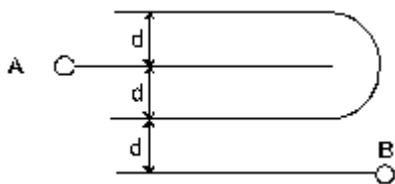


- (1) $\frac{7}{4}$ (2) $\frac{22}{3}$ (3) $\frac{3}{22}$ (4) $\frac{4}{7}$

45. A capacitor of capacitance C is initially charged to a potential difference of V volts. Now, it is connected to a battery of $2V$ emf with opposite polarity. The ratio of heat generated to the final energy stored in the capacitor will be :

- (1) 1.75 (2) 2.25
 (3) 2.5 (4) 1/2

46. Find out equivalent capacitance between A and B if each capacitor has capacitance ' C '.



- (1) $\frac{2C}{3}$ (2) $\frac{C}{3}$ (3) $\frac{4C}{3}$ (4) None

47. Magnetic field in a plane EM wave is given by $B_z = 2 \times 10^{-7} \sin(0.5 \times 10^3 x - 1.5 \times 10^{11} t)$ T then electric field is given by :-

- (1) $E_z = 60\sqrt{2} \sin(0.5 \times 10^3 x - 1.5 \times 10^{11} t)$ V/m
 (2) $E_z = 60 \sin(0.5 \times 10^3 x - 1.5 \times 10^{11} t)$ V/m
 (3) $E_y = 60\sqrt{2} \sin(0.5 \times 10^3 x - 1.5 \times 10^{11} t)$ V/m
 (4) $E_y = 60 \sin(0.5 \times 10^3 x - 1.5 \times 10^{11} t)$ V/m

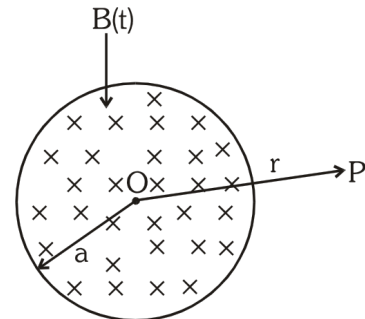
48. A cylindrical region of radius 1.2m is containing a uniform electric field that is increasing uniformly with time. At $t = 0$ the field is zero and at $t = 5.0$ s the field is 200 V/m. The total displacement current through a cross section of the region is :-

- (1) 4.5×10^{-16} A (2) 2.0×10^{-15} A
 (3) 3.5×10^{-10} A (4) 1.6×10^{-9} A

49. In a given series LCR circuit $R = 4\Omega$, $X_L = 5\Omega$ and $X_C = 8\Omega$, the current :-

- (1) Leads the voltage by $\tan^{-1}(3/4)$
 (2) Leads the voltage by $\tan^{-1}(5/8)$
 (3) Lags the voltage by $\tan^{-1}(3/4)$
 (4) Lags the voltage by $\tan^{-1}(5/8)$

50. A time varying magnetic field $B(t)$ exists in a circular region of radius ' a ' and is directed into the plane of the paper as shown in figure. The magnitude of the induced electric field at a point 'P' at a distance ' r ' from the centre of the circular region :-



- (1) $\frac{a^2}{2r} \left(\frac{dB}{dt} \right)$ (2) $\frac{a}{2r^2} \left(\frac{dB}{dt} \right)$
 (3) $\frac{1}{2r} \left(\frac{dB}{dt} \right)$ (4) $\frac{r}{2} \left(\frac{dB}{dt} \right)$

Topic : COORDINATION : FULL SYLLABUS EXCEPT IUPAC, p-BLOCK FULL EXCEPT FAMILIES, IONIC EQUILIBRIUM (TILL SALT HYDROLYSIS), [BACK UNIT] : GOC-II

SECTION-A (CHEMISTRY)

51. Magnetic moment of the complexes is zero in:-

- (1) $[\text{Ni}(\text{CN})_4]^{2-}$, $[\text{Ni}(\text{CO})_4]$
- (2) $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$, $[\text{Ni}(\text{CN})_4]^{2-}$
- (3) $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$, $[\text{Ni}(\text{CO})_4]$
- (4) $[\text{Fe}(\text{CN})_6]^{4-}$, $[\text{Fe}(\text{CN})_6]^{3-}$

52. The oxidation and coordination number of Pt in $[\text{Pt}(\text{C}_2\text{H}_4)\text{Cl}_3]^-$ is respectively :-

- (1) + 1, 3
- (2) + 2, 4
- (3) + 3, 6
- (4) + 2, 5

53. Select incorrect match for $[\text{M}(\text{H}_2\text{O})_6]^{+2}$ complex :-

	Metal Ion	Electronic configuration	CFSE
(1)	Mn^{+2}	$t_2g^3eg^2$	$0\Delta_0$
(2)	V^{+2}	$t_2g^3eg^0$	$-1.2\Delta_0$
(3)	Ni^{+2}	$t_2g^6eg^2$	$-1.6\Delta_0$
(4)	Ti^{+2}	$t_2g^2eg^0$	$-0.8\Delta_0$

54. The complex ions $[\text{Co}(\text{NH}_3)_5(\text{NO}_2)]^{2+}$ and $[\text{Co}(\text{NH}_3)_5(\text{ONO})]^{2+}$ are called :-

- (1) Ionization isomers
- (2) Linkage isomers
- (3) Coordination isomers
- (4) Geometrical isomers

55. Complex ion/compound having only two stereoisomers :

- (1) $[\text{CrBr}_2(\text{H}_2\text{O})(\text{NH}_3)_3]$
- (2) $[\text{Zn}(\text{gly})_2]$
- (3) $[\text{PtBrCl}(\text{CN})(\text{H}_2\text{O})]^-$
- (4) $[\text{Co}(\text{en})(\text{NH}_3)_4]^{3+}$

56. Which of the following species show facial and meridional isomer :

- (1) $[\text{Co}(\text{NH}_3)_3(\text{NO}_3)_3]$ (2) $[\text{Fe}(\text{H}_2\text{O})_3\text{Cl}_3]$
- (3) $[\text{Co}(\text{CN})_3\text{Br}_3]$ (4) All

57. Co-ordination number of Ni in $[\text{Ni}(\text{C}_2\text{O}_4)_3]^{4-}$ is :-

- (1) 3
- (2) 6
- (3) 4
- (4) 5

58. Which of the following is paramagnetic?

- (1) $\text{K}_4[\text{Fe}(\text{CN})_6]$ (2) $\text{K}_3[\text{Cr}(\text{CN})_6]$
- (3) $\text{K}_3[\text{Co}(\text{CN})_6]$ (4) $\text{K}_2[\text{Ni}(\text{CN})_4]$

59. The geometry with respect to central atom of the following molecules are :-

$\text{N}(\text{SiH}_3)_3$, $(\text{CH}_3)_3\text{N}$, $\text{P}(\text{SiH}_3)_3$

- (1) Planar, pyramidal, planar
- (2) Planar, Pyramidal, pyramidal
- (3) Pyramidal, Pyramidal, Pyramidal
- (4) Pyramidal, planar, Pyramidal

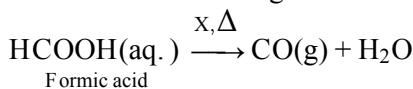
60. Which of following will not exist :-

- (1) NF_5 (2) BiI_5 (3) PbI_4 (4) All

61. Hydrolysis of PCl_5 gives :-

- (1) H_3PO_2 (2) H_3PO_3 (3) H_3PO_4 (4) HOCl

62. Consider the following chemical reaction



In the above reaction (X) is –

- (1) Conc. H_2SO_4 (2) Conc. HCl
- (3) Conc. HNO_3 (4) Dil. HNO_3

63. Silicate having one monovalent corner oxygen atom in each tetrahedron unit is:

- (1) sheet silicate
- (2) cyclic silicate
- (3) single chain silicate
- (4) double chain silicate

64. Which of the following molecule is having complete octet ?
 (1) BeCl_2 (dimer) (2) BeH_2 (dimer)
 (3) BeH_2 (s) (4) BeCl_2 (s)
65. Which of the following complex ion is expected to absorb visible light ?
 (1) $[\text{Ni}(\text{H}_2\text{O})_6]^{+2}$ (2) $[\text{Cr}(\text{NH}_3)_6]^{+3}$
 (3) $[\text{Fe}(\text{H}_2\text{O})_6]^{+2}$ (4) All
66. The type of isomerism shown by the complex $[\text{Fe}(\text{en})_2\text{Cl}_2]$
 (1) Geometrical (2) Optical
 (3) 1 & 2 (4) Linkage
67. Which of the following carbonyls will have the longest C-O bond length ?
 (1) $[\text{Fe}(\text{CO})_5]$ (2) $[\text{Mn}(\text{CO})_6]^+$
 (3) $[\text{Cr}(\text{CO})_6]$ (4) $[\text{V}(\text{CO})_6]^-$
68. Which of the following complex will exhibit highest paramagnetic behaviours.
 (1) $[\text{Ti}(\text{NH}_3)_6]^{+3}$
 (2) $[\text{Co}(\text{CN})_6]^{-3}$
 (3) $[\text{Zn}(\text{NH}_3)_6]^{+2}$
 (4) $[\text{Sc}(\text{H}_2\text{O})_3(\text{NH}_3)_3]^{+3}$
69. The stability of +3 oxidation state among group 13 elements will be ?
 (1) $\text{B} > \text{Al} < \text{Ga} > \text{In} < \text{Tl}$
 (2) $\text{B} < \text{Al} < \text{Ga} < \text{In} > \text{Tl}$
 (3) $\text{B} > \text{Al} > \text{Ga} > \text{In} < \text{Tl}$
 (4) $\text{B} > \text{Al} > \text{Ga} > \text{In} > \text{Tl}$
70. Which of the following is used as dry bleach ?
 (1) H_2O_2 (2) Moist SO_2
 (3) O_3 (4) Both 1 & 3
71. At infinite dilution a weak electrolyte contains:
 (1) Unionised molecules
 (2) Only ions
 (3) Ions and unionised molecules
 (4) Not definite
72. The pK_a of HNO_2 is 3.37. The pH of HNO_2 in its 0.01 mol L^{-1} aqueous solution will be :-
 (1) 5.37 (2) 2.69
 (3) 1.69 (4) 0.69
73. At 80°C , Pure distilled water has H_3O^+ concentration equal to $1 \times 10^{-6} \text{ mol L}^{-1}$. The value of K_w at this temperature will be :-
 (1) 1×10^{-8} (2) 1×10^{-12}
 (3) 1×10^{-14} (4) 10^{-6}
74. On increasing temperature which of the following is not correct for water?
 (1) degree of ionization increases
 (2) K_w increases
 (3) $[\text{H}^+]$ increases
 (4) pH increases
75. 100 ml of 0.2 M H_2SO_4 is added to 100 ml of 0.2 M NaOH. The resulting solution will be :-
 (1) Acidic
 (2) Basic
 (3) Neutral
 (4) Slightly basic
76. Which of the following is the electrophile that attacks the aromatic ring during friedel crafts acylation ?
 (1) $\text{R}-\text{C}(=\text{O})-\text{Cl}$
 (2) $\text{R}-\text{C}^+=\text{O}$
 (3) $\text{R}-\text{C}(=\text{O})-\text{Al}^+\text{Cl}_2$
 (4) $\text{R}-\text{C}(\text{O})-\text{Cl}-\text{Cl}^+$

77. Which of the following compound gives acetaldehyde with $\text{HOH}/\text{H}_2\text{SO}_4/\text{HgSO}_4$?

- (1) $\text{CH}\equiv\text{CH}$ (2) $\text{CH}_3-\text{C}\equiv\text{CH}$
(3) $\text{CH}_2=\text{CH}_2$ (4) $\text{CH}_3-\text{CH}=\text{CH}_2$

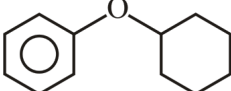
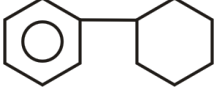
78. $\text{CH}_3-\text{CH}_2-\overset{\text{O}}{\underset{\text{O}}{\text{C}}}-\text{H} \xrightarrow{\text{Red P + HI}}$ Product (A)

Find the product (A) of following reaction.

- (1) $\text{CH}_3-\text{CH}_2-\text{COOH}$
(2) $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{OH}$
(3) $\text{CH}_3-\text{CH}_2-\text{CH}_3$
(4) $\text{CH}_3-\overset{\text{O}}{\underset{\text{O}}{\text{C}}}-\text{CH}_3$

79. $\text{C}_6\text{H}_6 + \text{C}_6\text{H}_{11}\text{OH} \xrightarrow{\text{MgBr}}$?

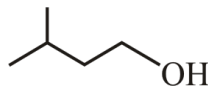
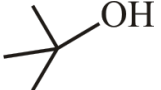
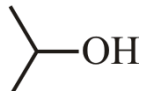
Product is ?

- (1) 
(2) 
(3) 
(4) 

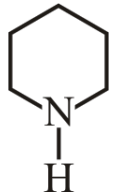
80. $\text{CH}_3-\text{COOH} \xrightarrow{\text{SOCl}_2} (\text{A}) \xrightarrow[\text{Anhyd. AlCl}_3]{\text{Benzene}}$ (B) product is

- (1) $\text{CH}_3-\overset{\text{O}}{\underset{\text{O}}{\text{C}}}-\text{CH}_3$
(2) $\text{CH}_3-\overset{\text{O}}{\underset{\text{O}}{\text{C}}}-\text{C}_6\text{H}_5$
(3) $\text{C}_6\text{H}_5-\overset{\text{O}}{\underset{\text{O}}{\text{C}}}-\text{C}_6\text{H}_5$
(4) $\text{CH}_3-\overset{\text{O}}{\underset{\text{O}}{\text{C}}}-\text{H}$

81. In Victor Meyer's test red colour is given by :-

- (1)  (2) 
(3)  (4) 

82. Which of the following give positive iso cyanide test ?

- (1) 
(2) $\text{CH}_3-\text{NH}-\text{CH}_3$
(3) 
(4) $\text{CH}_3-\text{CH}_2-\text{NH}_2$

83. $\text{CH}_3-\text{CH}_2-\text{C}\equiv\text{N} \xrightarrow{(\text{x})} \text{CH}_3-\text{CH}_2-\text{CHO}$
The reagent 'x' is :-

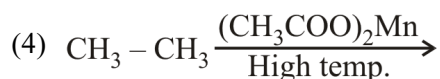
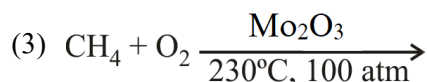
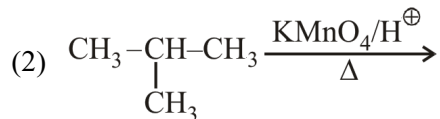
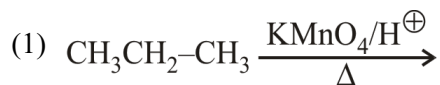
- (1) $\text{SnCl}_2/\text{HCl}, \text{H}_3\text{O}^+$
(2) $\text{H}_2/\text{Pd}-\text{BaSO}_4$
(3) $\text{LiAlH}_4/\text{H}_3\text{O}^+$
(4) NaBH_4

84. $\text{C}_6\text{H}_6 + \text{CH}_3-\overset{\text{O}}{\underset{\text{O}}{\text{C}}}-\text{Cl} \xrightarrow{\text{Anhyd. AlCl}_3} (\text{A}) \xrightarrow{(\text{x})} \text{C}_6\text{H}_5-\text{CH}_2\text{CH}_3$

Reagent (x) can be :-

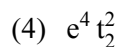
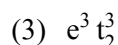
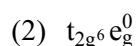
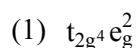
- (1) Red P + HI
(2) Zn-Hg/HCl
(3) $\text{NH}_2-\text{NH}_2 / \text{OH}^-$
(4) All of the above

85. Which of the following oxidation reaction of alkane is not possible ?

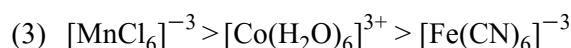
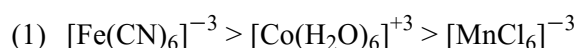


SECTION-B (CHEMISTRY)

86. What is the correct electronic configuration of the central atom in $\text{Na}_4[\text{FeF}_6]$ based on CFT ?



87. Which one of the following is the correct order of the wavelength absorption in the visible region for the given complexes ?



88. Which of these statements about $[\text{Ni}(\text{CO})_4]$ is true :

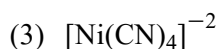
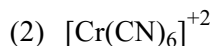
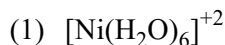
(1) dsp^2 hybridisation & has no unpaired electrons

(2) sp^3 hybridisation & has 2 unpaired electrons

(3) dsp^2 hybridisation & has 2 unpaired electrons

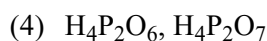
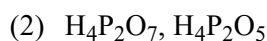
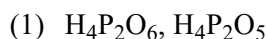
(4) None of these

89. A magnetic moment of 2.83 will be shown by :



(4) 1 & 2

90. In which pair of ions both the species contain P-O-P bond ?



91. Which of the following statement is not correct about diborane ?

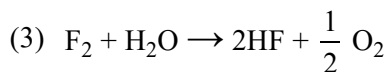
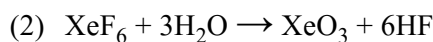
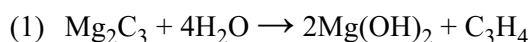
(1) Both the boron atoms are sp^3 hybridised

(2) There are two, 3-centre 2-electron bonds.

(3) The four terminal H-atoms and 2 boron atoms in the same plane

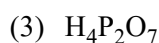
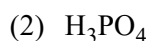
(4) It is planar in structure

92. Which of the following does not come under hydrolysis type reaction ?



(4) All

93. Which of the following oxyacids has basicity equal to number of H atoms present ?



(4) All

94. How many H^+ ions are present in 1 ml of a solution whose pH is 13 ?

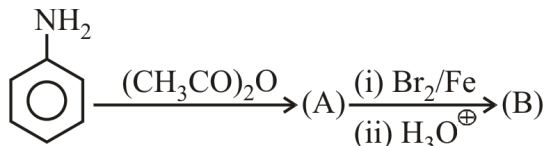
- (1) 10^{-16} (2) 6.022×10^{13}
(3) 6.022×10^7 (4) 6.022×10^{23}

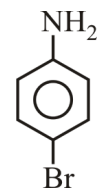
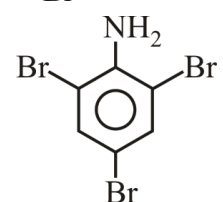
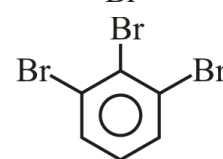
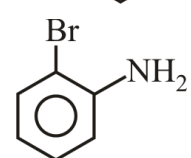
95. Which of the following is the weakest acid ?

- (1) Phenol ($K_a = 1.3 \times 10^{-10}$)
(2) Hydrocyanic acid ($K_a = 4.9 \times 10^{-10}$)
(3) Acetic acid ($K_a = 1.8 \times 10^{-5}$)
(4) Benzoic acid ($K_a = 6.5 \times 10^{-5}$)

96. $CH_3-\underset{\substack{| \\ F}}{CH}-CH_2-CH_3 \xrightarrow[\Delta]{Alic. KOH}$ Major product
Incorrect statement for major product ?

- (1) Show G.I.
(2) Product is less stable than minor product.
(3) Product shows hyper conjugation.
(4) Product is unsaturated.

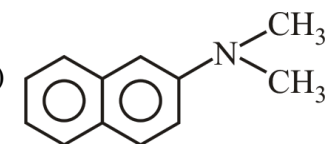
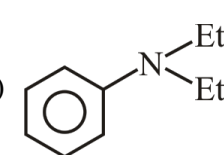
97. 
major product (B) is :-

- (1) 
(2) 
(3) 
(4) 

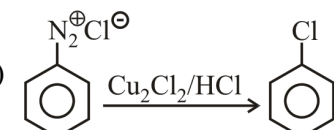
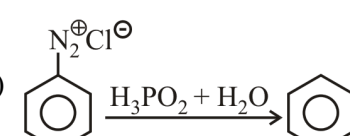
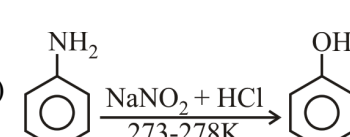
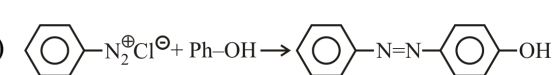
98. Which of the following is incorrect ?

- (1) $CH_3-CN \xrightarrow[(2) H_3O^+]{(1) LiAlH_4} CH_3-CH_2-NH_2$
(2) $CH_3-NC \xrightarrow[(2) H_3O^+]{(1) LiAlH_4} CH_3-NH-CH_3$
(3) $CH_3-CONH_2 \xrightarrow[(2) H_3O^+]{(1) LiAlH_4} CH_3-CH_2-OH$
(4) $CH_3-CONH_2 \xrightarrow{Br_2 + KOH} CH_3-NH_2$

99. Which of the following amine will be soluble in alkali during Hinsberg's test.

- (1) 
(2) $CH_3-CH_2-NH_2$
(3) $(C_2H_5)_2NH$
(4) 

100. Which of the following is incorrect ?

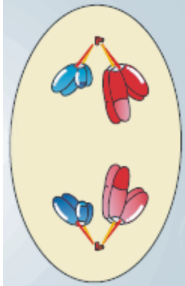
- (1) 
(2) 
(3) 
(4) 

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Topic : SEX LINKAGE TO POPULATION GENETICS, BIOTECH, PTC, PLANT BREEDING, MICROBES, CELL DIVISION, PHOTOSYNTHESIS IN HIGHER PLANT & ENZYME, NERVOUS SYSTEM, EYE, [BACK UNIT] : GENETICS (MODULE 9 & 10)

SECTION-A (BIOLOGY-I)

101.



Choose the correct statement for given figure :-

- (1) Chromosome are move to spindle equator
 - (2) Centromere split and chromatids seprate
 - (3) Golgicomplex and ER reform
 - (4) Homologous chromosomes separate
102. If in a cell 28 bivalents are present in prophase-I, then what will be the no. of chromosomes in Anaphase-II.
- (1) 28
 - (2) 14
 - (3) 56
 - (4) 112
103. Read the following statements and identify them as true (T) or false (F) and accordingly choose correct option.
- A.** Splitting of centromere takes place during anaphase of mitosis.
- B.** During metaphase of mitosis each chromatid of a chromosome remain connected to pole of its side by spindle fibre attached at its kinetochore
- C.** Nucleolus, Golgi complex and ER reform during end of prophase.
- D.** In animal cells cytokinesis is carried out by furrow formation.
- Options :-**
- | | A | B | C | D |
|-----|---|---|---|---|
| (1) | T | T | F | F |
| (2) | T | T | F | T |
| (3) | F | T | T | T |
| (4) | T | F | F | T |
104. At which stage of meiosis, sister chromatids do not separate and attached to centromere?
- (1) Anaphase-II
 - (2) Anaphase-I
 - (3) Metaphase-II
 - (4) Both (2) & (3)
105. S-phase is not characterized by :-
- (1) DNA duplication
 - (2) DNA replication
 - (3) No increase in chromosome number
 - (4) Increase in chromosome number
106. Experiment using two leaves a variegated leaf or a leaf that was covered with black paper and one that was exposed to light, on testing these leaves for starch shows that photosynthesis occurs only in _____ part of leaf in _____ of sun light.
- (1) non-green, presence
 - (2) green, presence
 - (3) non-green, absence
 - (4) green, absence
107. The first acceptor of $^{14}\text{CO}_2$ in rice plant during carbon fixation cycle is ?
- (1) PEP
 - (2) Malic acid
 - (3) RuBP
 - (4) PGA
108. Fixation of 12CO_2 in C_3 -cycle needs :-
- (1) $12 \text{ NADPH} + \text{H}^+$, 12 ATP
 - (2) $12 \text{ NADPH} + \text{H}^+$, 18 ATP
 - (3) $24 \text{ NADPH} + \text{H}^+$, 36 ATP
 - (4) $36 \text{ NADPH} + \text{H}^+$, 36 ATP

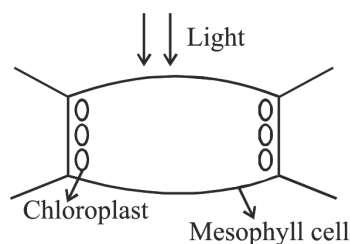
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109.



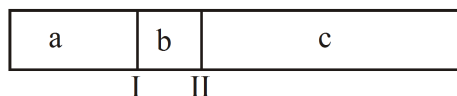
In the given above diagram, orientation of chloroplast in a mesophyll cell is shown. Which of the following is correct about light ?

- (1) Low intensity light
 - (2) High intensity light
 - (3) Moderate intensity light
 - (4) Nothing can be predicted on the basis of given data
110. Kranz anatomy found in :-
- (1) C_3 plant
 - (2) C_4 plant
 - (3) CAM plant
 - (4) C_2 plant
111. Generally enzymes are different from inorganic catalyst because :-
- (1) Not being used up in reaction
 - (2) Change the rate of reaction
 - (3) Having high diffusion rate
 - (4) Can not work at high temperature
112. Which one of the following statements is correct, with reference to enzymes ?
- (1) Holoenzyme = Apoenzyme + Co-factor
 - (2) Coenzyme = Apoenzyme + Holoenzyme
 - (3) Holoenzyme = Coenzyme + Co-factor
 - (4) Apoenzyme = Holoenzyme + Coenzyme
113. In photosynthesis photolysis of water is concerned with OEC. This OEC is connected to :-
- (1) PS-II only
 - (2) PS-I only
 - (3) PS-I and PS-II both
 - (4) NADH-reeducates enzyme.

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114. How many chlorophyll-a molecules form the reaction center in photosystem-I ?
- (1) Many
 - (2) One
 - (3) Three
 - (4) Six
115. The stage in which cell is metabolically active but fails to divide is :-
- (1) G_1 phase
 - (2) G_0 phase
 - (3) S-phase
 - (4) M-phase
116. Non protein organic substance which can not be easily separated from apo enzyme is :-
- (1) Prosthetic group
 - (2) Co-enzyme
 - (3) Holoenzyme
 - (4) Activator
117. Haem act as prosthetic group for :
- (1) Carboxypeptidase
 - (2) Pepsin and Trypsin
 - (3) Peroxidase & Catalase
 - (4) Urease & Kinase
118. Given below are two statements :
- Statement I :** RNA mutates at a faster rate
- Statement II :** Viruses having RNA genome and shorter life span mutate and evolve faster
- In the light of the above statements choose the correct answer from the options given below
- (1) Both Statements I and Statement II are false
 - (2) Statement I is true but Statement II is false
 - (3) Statement I false but Statement II is true
 - (4) Both Statement I and Statement II are true
119. Real Inducer of the lac operon is :
- (1) Lactose
 - (2) Allolactose
 - (3) Glucose
 - (4) Galactose

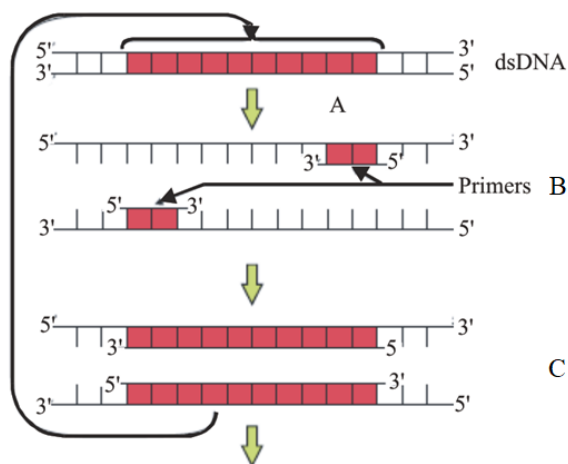
130. The given segment of DNA has restriction sites I and II, which create restriction fragments a, b and c :



Select the correct diagram of gel :-

- (1) - +
- (2) - +
- (3) - +
- (4) - +

131. The below diagram refer to PCR. Identify the steps A, B and C and select the correct option.



- (1) A-Denaturation at 94°C, B-Annealing at 54°C, C-Extension through taq polymerase at 72°C
- (2) A-Annealing at 94°C, B-Denaturation at 54°C, C-Extension through taq polymerase at 72°C
- (3) A-Extension through taq polymerase at 54°C, B-Denaturation at 94°C, C-Annealing at 94°C
- (4) A-Extension through taq polymerase at 72°C, B-Denaturation at 94°C, C-Annealing at 54°C

132. **Assertion** : Polymerase chain reaction is the technique that serve the purpose of early diagnosis.

Reason : Polymerase chain reaction based on the principle by amplification of their nucleic acid.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

133. **Assertion** :- In order to force bacteria to take up the plasmid, the bacteria cells must first be made 'competent' to take up DNA.

Reason :- DNA is a hydrophilic molecule, it cannot pass through cell membranes.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

134. If recombinant DNA is inserted within the coding sequence of enzyme galactosidase, which of the following will occur in case of non-recombinants:-

- (1) Insertional inactivation
- (2) Colonies do not produce any colour
- (3) Chromogenic substrate gives blue colour
- (4) Inactivation of enzyme galactosidase

135. Insertional inactivation of selectable marker genes is helpful in selecting the :-

- (1) Transformants
- (2) Non-transformants
- (3) Recombinants
- (4) Non-recombinants

SECTION-B (BIOLOGY-I)

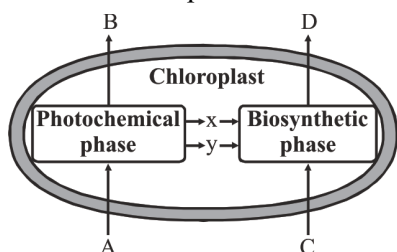
136. During cell cycle DNA polymerase activity and histon protein synthesis occurs in -

- (1) G_0
- (2) G_1
- (3) S
- (4) G_2

137. Find the incorrect statement :

- (1) In bundle sheath cells of C_4 plants, first product of CO_2 fixation is 3C compound.
- (2) In photorespiration both chloroplast and peroxisome utilize O_2 .
- (3) PEP carboxylase plays an important role in synthesis of sugar in mesophyll cells of C_4 plants.
- (4) ATP requirement of C_4 plants is higher as compared to C_3 plants to produce one molecule of sugar.

138. The diagram shows the movement of substances in and out of the chloroplast :-



What do labels A to D represent ?

	A	B	C	D
(1)	H_2O	Sugar	H_2O	O_2
(2)	CO_2	H_2O	Sugar	O_2
(3)	Sugar	H_2O	CO_2	Starch
(4)	H_2O	O_2	CO_2	Carbohydrate

139. The sensitivity of C_4 plants to low temperature than C_3 plant is due to -

- (1) High CO_2 compensation point
- (2) Pyruvate phosphate dikinase
- (3) Kranz anatomy
- (4) Pyruvate kinase

140. Which one of the following options gives the correct match of processes (A, B, C) with the products they give out ?

	A Z-scheme of light reaction	B Photorespiration	C Biosynthetic phase of C_4 plants
(1)	Glucose, ATP and oxygen	ATP and NADPH	ATP, oxygen and sucrose
(2)	ATP, NADPH and oxygen	Phosphoglycerate	Sucrose
(3)	Only ATP	Only NADPH	Only oxygen
(4)	Water, NADPH and CO_2	Phosphoglycerate and ATP	Glucose, fructose and sucrose

141. The correct sequence of cell cycle is :-

- (1) $G_1 \rightarrow G_2 \rightarrow S \rightarrow M$
- (2) $G_1 \rightarrow S \rightarrow G_2 \rightarrow M$
- (3) $M \rightarrow G_2 \rightarrow S \rightarrow G_1$
- (4) $S \rightarrow G_1 \rightarrow M \rightarrow G_2$

142. Longest phase of cell division is :-

- (1) Prophase
- (2) Anaphase
- (3) Metaphase
- (4) Telophase

143. If at the end of meiosis each daughter cell has four chromosomes, how many chromosomes would a diploid cell have in G_1 phase of cell cycle.

- (1) 2
- (2) 4
- (3) 8
- (4) 16

144. Genetic code is

- (1) triplet, universal, ambiguous and degenerate
- (2) triplet, universal, non-ambiguous and non degenerate
- (3) triplet, universal, non-ambiguous and degenerate
- (4) triplet, universal, ambiguous and non-degenerate

145. Basis of DNA fingerprinting is :-

- (1) high degree of polymorphism in non coding sequences
- (2) low degree of polymorphism in non coding sequences
- (3) intermediate degree of polymorphism in coding sequences
- (4) sequences showing no polymorphism

146. The first letter of the name of restriction endonuclease came from the :-

- (1) genus of organism
- (2) species of organism
- (3) family of organism
- (4) class of organism

147. 'Restriction' in Restriction enzyme refers to:-

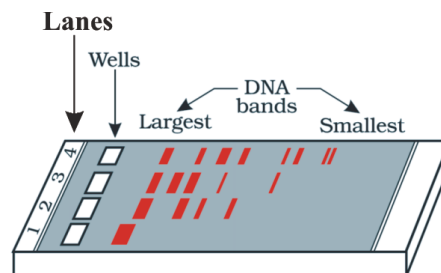
- (1) Cleaving of phosphodiester bonds in DNA
- (2) Cutting of DNA at any site
- (3) Prevention of multiplication of bacteriophage in bacteria
- (4) All of the above

148. In gel electrophoresis, the separated DNA fragments are visualised after staining the DNA with ...A... followed by exposure to ...B... Here

A and B refers to :-

	A	B
1	Ethidium bromide	Infrared radiation
2	Ethidium bromide	UV radiation
3	Ethidium nitrate	γ -rays
4	Ethidium chloride	Radiowave

149. In following figure of gel electrophoresis which lane is showing undigested set of DNA sample ?



- (1) lane - 1
- (2) lane - 2
- (3) lane - 3
- (4) lane - 4

150. If a pBR322 vector is cut by Pst I enzyme then which of its marker gene shows insertional inactivation-

- (1) Tet^R gene
- (2) Amp^R gene
- (3) Gal gene
- (4) Rop gene

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Topic : SEX LINKAGE TO POPULATION GENETICS, BIOTECH, PTC, PLANT BREEDING, MICROBES, CELL DIVISION, PHOTOSYNTHESIS IN HIGHER PLANT & ENZYME, NERVOUS SYSTEM, EYE, [BACK UNIT] : GENETICS (MODULE 9 & 10)

SECTION-A (BIOLOGY-II)

151. Select the incorrect statement for continuous culture system :-

- (1) In this used medium is drained out from one side while fresh medium is added from other side.
- (2) In this cells are maintained in their physiologically less active lag phase of growth.
- (3) It produces larger biomass
- (4) It shows higher yields of desired product.

152. PCR process requires :-

- (1) Two sets of DNA primers
- (2) Thermosensitive DNA polymerase
- (3) ATP, GTP, CTP, TTP
- (4) All of the above

153. **Assertion (A)** :- RNAi involves silencing of a specific mRNA due to complementary RNA that binds to & prevents translation of the mRNA.

Reason (R) :- RNAi takes place in all prokaryotic organisms as a method of cellular defense.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

154. Genetically modified golden rice variety where a gene of ____?____ was transferred :-

- (1) β - carotene
- (2) β - galactose
- (3) Vitamin B
- (4) α - carotene

155. Which of the following techniques serve the purpose of early diagnosis ?

A – rDNA Technology

B – PCR

C – ELISA

- (1) A only
- (2) A & C only
- (3) A & B only
- (4) A, B & C

156. Variety of bhindi that shows resistance to yellow mosaic virus :-

- (1) Prabhani Kranti
- (2) Pusa sawani
- (3) Pusa komal
- (4) Himgiri

157. Breeding crops with higher level of vitamins and mineral, higher protein and healthier fat is :-

- (1) Bioremediation
- (2) Biofortification
- (3) Bio energy
- (4) Bio fuel

158. Match the following :-

	A		B
i	Somaclones	a	Protoplast of different varieties of plants are fused
ii	Totipotency	b	Capacity to generate a whole plant from any plant cell
iii	Somatic hybrid	c	Genetically identical to original plant
iv	Meristem culture	d	Virus free plant

Option:-

- (1) i-a, ii-b, iii-c, iv-d
- (2) i-c, ii-b, iii-a, iv-d
- (3) i-c, ii-a, iii-d, iv-b
- (4) i-d, ii-c, iii-b, iv-a

159. Biological method of control of pests relies on:-

- (1) Introduced chemicals
- (2) Natural predation
- (3) Intraspecific competition
- (4) Biofertilizers

160. Microbes like Spirulina, Methylophilus methylotrophus can be grown on industrial scale as sources of good:-

- (1) fat
- (2) carbohydrate
- (3) protein
- (4) All of these

161. **Assertion (A)** → Farming of spirulina also reduces environmental pollution.

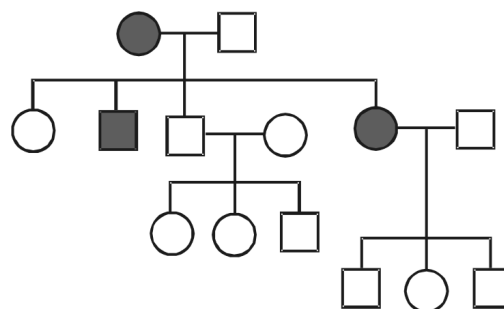
Reason (R) → Spirulina can be grown on waste water, straw molasses, starch, animal manure and even sewage

- (1) A & R correct, R is correct explanation.
- (2) A & R correct, R is not correct explanation.
- (3) A is correct, R is incorrect
- (4) Both A and R incorrect

162. Which of the following usually do not form polymers?

- (1) Amino acids
- (2) Nucleotides
- (3) Lipids
- (4) Monosaccharides

163. Study the pedigree chart of a family showing the inheritance of myotonic dystrophy :-



The trait under study is

- (1) Dominant X-linked
- (2) Recessive X-linked
- (3) Autosomal dominant
- (4) Recessive Y-linked

164. Mendel did not observe linkage due to:-

- (1) Mutation
- (2) Synapsis
- (3) Crossing over
- (4) independent assortment

165. Which of the following crosses will give tall and dwarf pea plants in same proportions ?

- (1) $TT \times tt$
- (2) $Tt \times tt$
- (3) $TT \times Tt$
- (4) $tt \times tt$

166. In the Punnett square given below, the genotype of female parent will be ?

		Male gametes			
Female gametes	?	AB	Ab	aB	ab
		AABb	AAbb	AaBb	Aabb

- (1) AaBb
- (2) AAbb
- (3) aaBB
- (4) Aabb

167. In pea plant which of the following character is controlled by pleiotropic gene :-

- (1) Height and flower colour
- (2) Seed shape and size of starch grain
- (3) Seed colour and pod colour
- (4) Flower colour and seed shape

168. In case of polygenic inheritance, if a trait is controlled by four gene, then in F_2 -generation, what will be the number of dominant alleles in the organism that obtained in maximum number:-

- (1) 8
- (2) 3
- (3) 4
- (4) 5

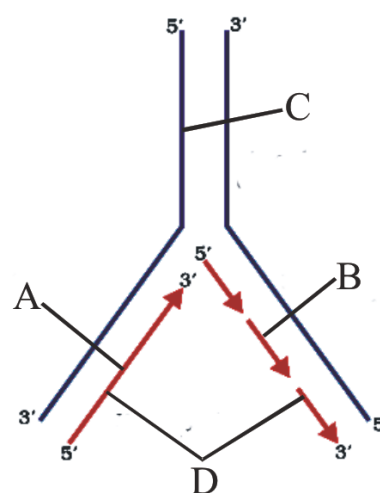
169. Identify incorrect match :-

- (1) α -1-antitrypsin - Emphysema
- (2) Genetically engineered insulin - Humulin
- (3) Myocardial infarction - ELISA
- (4) 1st Transgenic cow - Rosie

170. The packaging of chromatin in eukaryotes at higher level requires set of proteins that are collectively referred to as :-

- (1) Histones
- (2) Polyamines
- (3) Non-Histone Chromosomal protein
- (4) Globins

171. Identify part A, B, C, D in the figure given below:-



	A	B	C	D
(1)	Continuous synthesis	Discontinuous synthesis	Template strand	Newly synthesised strands
(2)	Discontinuous synthesis	Continuous synthesis	Newly synthesised strand	Template strand
(3)	Template strand	Newly synthesised strand	Continuous synthesis	Discontinuous synthesis
(4)	Newly synthesised strand	Template strand	Continuous synthesis	Discontinuous synthesis

172. Match the column-I with column-II and identify the correct option :-

Column-I		Column-II	
A.	Axoplasm	1.	Impermeable to -ve charged protein
B.	Axolemma	2.	Low K^+ high Na^+
C.	Fluid outside axon	3.	Active transport
D.	$Na^+ K^+$ pump	4.	High K^+ and low Na^+

A B C D

- (1) ii i iv iii
 (2) ii iv i iii
 (3) iv i ii iii
 (4) iv ii i iii

173. If the $Na^+ - K^+$ pump stops working then the

- (1) Na^+ and K^+ will be in excess in extracellular fluid
 (2) Na^+ will be in excess in extracellular fluid
 (3) K^+ will be excess in intracellular fluid
 (4) Na^+ will be in excess in intracellular fluid

174. Match the column-A and column-B :-

	Column-A		Column-B
(a)	Cerebral hemispheres	(i)	Intersensory association
(b)	Corpus callosum	(ii)	Urge for eating & drinking
(c)	Association areas	(iii)	Connect two hemispheres
(d)	Hypothalamus	(iv)	Cerebrum divided into two halves

Choose the correct match for above :-

- (1) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)
 (2) (a)-(ii), (b)-(iii), (c)-(i), (d)-(iv)
 (3) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)
 (4) (a)-(iii), (b)-(iii), (c)-(i), (d)-(iv)

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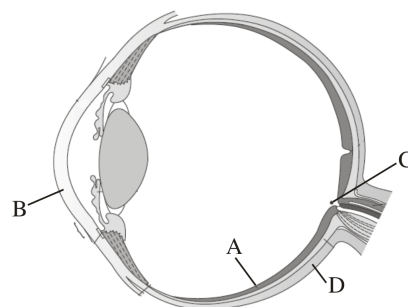
175. Which of the following statement is false :-

- (1) Midbrain and hind brain form the brain stem.
 (2) Pons consists of fibre tracts that interconnect different regions of the brain.
 (3) The neuron cell bodies are concentrated in the corpus callosum.
 (4) The layer of cell bodies which covers the cerebral hemisphere is called cerebral cortex and is thrown into prominent folds.

176. Identify wrongly matched pair of a structure and its location :-

	Structure	Location
1.	Neurofibril	Axon, cyton, Dendron
2.	Nissl's granule	Dendron, cyton
3.	Synaptic vesicles	Near nucleus in cyton
4.	Ligand gated channels	Dendron

177. Parts A, B, C, D of human eye are shown in the diagram. Select the option which gives correct identification along with its functions :-



- (1) D - Sclera - its anterior part forms ciliary body.
 (2) A - Retina - it has bipolar neurons
 (3) C - Blind spot - has only rods
 (4) B - Cornea - pigmented visible portion of choroid

178. Association areas as :-

- (1) Sensory in function.
- (2) Motor in function.
- (3) Neither clearly sensory nor motor in function.
- (4) Neurosecretory in function.

179. Which part of the brain is a major coordinating centre for sensory and motor signaling ?

- (1) Association area
- (2) Thalamus
- (3) Hypothalamus
- (4) Limbic system

180. A mammalian brain is characterised by the presence of :-

- (1) Pons
- (2) Corpus callosum
- (3) Cerebellum
- (4) Cerebrum

181. Broca's area is connected with :-

- (1) Sensation of smell
- (2) Learning and reasoning
- (3) Speech
- (4) Receiving impulse from eye

182. Which one does not involve brain ?

- (1) Spinal reflex
- (2) Control of ANS
- (3) Cranial reflex
- (4) Voluntary action

183. Which is the visible coloured portion of eye ?

- (1) Cornea
- (2) Iris
- (3) Lens
- (4) Retina

184. The anterior portion on of the sclera is called :-

- (1) Cornea
- (2) Pupil
- (3) Ciliary body
- (4) Iris

185. Function of Iris is to :-

- (1) Move lens forward and backward.
- (2) Refract light rays.
- (3) Bring about movements of eyelids.
- (4) Alter the size of pupil.

SECTION-B (BIOLOGY-II)

186. The proteins encoded by the gene cryIAb control the....A.... and cryIIAb control the....B....

Choose the correct word for I and II.

	A	B
1	cotton bollworm	tabacco budworm
2	corn borer	cotton bollworm
3	cotton bollworm	corn borer
4	tabacco budworm	corn borer

187. Crystals of Bt toxin produced by some bacteria do not kill the bacteria themselves because :

- (1) Toxin is inactive
- (2) Toxin is immature
- (3) Bacteria are resistant to the toxin
- (4) Bacteria encloses toxin in a special sac

188. Gene therapy first used in the treatment of :-

- (1) Phenylketonuria
- (2) Albinism
- (3) SCID
- (4) Haemophilia

189. 'Pomato' is developed by which tissue culture method :-

- (1) Meristem culture
- (2) Embryo culture
- (3) Anther culture
- (4) Somatic hybridisation

190. Virus free plants can be obtained by

- (1) Only apical meristem
- (2) Only axillary meristem
- (3) Apical and axillary meristem
- (4) Embryo culture

191. Biological active form of protein is :-

- (1) Primary structure
- (2) Secondary structure
- (3) Tertiary structure
- (4) All of the above

192. About genotype Tt select out the incorrect statement:-

- (1) Allelic gene pair
- (2) Can be present on non homologous chromosome
- (3) They share common locus
- (4) They share same character

193. Four couples are claiming to be the parents of a child having 'O' blood group. Which of the following couples' claim is baseless, with blood groups of male and female parents, respectively as :-

- (1) A and B
- (2) A and O
- (3) B and O
- (4) AB and O

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194. How many enzymes/proteins are related to DNA replication process :

RNA ligase, RNA polymerase, Release factor, Helicase, DNA polymerase, DNA ligase, Polynucleotide phosphorylase

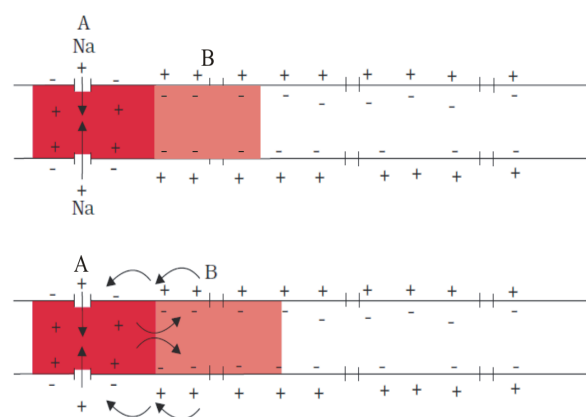
- (1) Eight (2) Four (3) Seven (4) Five

195. **Assertion:** Within a fraction of a second, resting potential of the membrane of Nerve restore.

Reason: The rise in the stimulus induced permeability to Na^+ is extremely short lived, it is quickly followed by a rise in a permeability to K^+

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

196. Find direction respectively (i) conduction of nerve impulse (ii) flow of current on outside the axolemma in given diagram.



- (1) (i) $A \rightarrow B$, (ii) $B \rightarrow A$
- (2) (i) $A \rightarrow B$, (ii) $A \rightarrow B$
- (3) (i) $B \rightarrow A$, (ii) $B \rightarrow A$
- (4) (i) $B \rightarrow A$, (ii) $A \rightarrow B$

197. Read the following lines and identify the part of brain :-

- (a) It is the major coordinating centre for sensory and motor signalling
- (b) It forms 80% of diencephalon
- (c) It does not secrete any hormone

- (1) Cerebellum
- (2) Hypothalamus
- (3) Thalamus
- (4) Corpora quadrigemina

198. The arrangement of retinal layer from outside to inside is :-

	Outer	Middle	Inner
(1)	Ganglionic cells	Bipolar cells	Photoreceptor cells
(2)	Photoreceptor cells	Bipolar cells	Ganglion cells
(3)	Bipolar cells	Photoreceptor cells	Ganglion cells
(4)	Bipolar cells	Ganglion cells	Photoreceptor cells

199. The forebrain consists of :-

- (1) Cerebellum, thalamus and hypothalamus.
- (2) Pons, cerebrum and medulla.
- (3) Hypothalamus, cerebrum and thalamus.
- (4) Amygdala, hippocampus and brain stem.

200. The point of eye where the visual acuity (resolution) is greatest :-

- (1) Blind spot
- (2) Macula lutea
- (3) Fovea centralis
- (4) Corpus lutea

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